SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO5 – Articulation	**Assessment:**	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	B.TECH (AI & DS)	**Date:**	

Code of Conduct

I VIVETHA B 192424170 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: VIVETHA B

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

Annexure A – Constraint-Based Problem Solving

1. **Given the following paragraph:**
   "John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

Tasks: a) Identify all referring expressions (mentions) and possible antecedents.

b) Apply the following **constraints** to resolve the references:

- Gender and number agreement
- Recency (prefer recent antecedents)
- Semantic compatibility (e.g., animacy and action suitability)
- Coherence (maintain consistent entity chains)

c) Draw a **constraint graph** or table showing variables, domains, and which constraints eliminate possible links.

d) Provide the final resolved coreference chains and rewrite the paragraph with clear references.

e) Justify the priority order of your constraints and discuss what would happen if the recency constraint is relaxed.

a) Referring expressions and possible antecedents

The paragraph contains several **referring expressions (mentions)** such as pronouns and noun phrases that refer to previously introduced entities.

1. **John** — introduces entity **John**
2. **Mary** — introduces entity **Mary**
3. **the park** — introduces entity **park**
4. **He** — possible antecedents: John, Mary
5. **a ball** — introduces entity **ball**
6. **She** — possible antecedents: John, Mary
7. **it** — possible antecedents: ball, park
8. **The dog** — introduces entity **dog**
9. **him** — possible antecedents: John, Mary, dog
10. **they** — possible antecedents: John, Mary, dog; possibly John and Mary as a pair

b) Applying the constraints

1. Gender and number agreement

- **He → John**
  - John is masculine and singular.
  - Mary is feminine, so **He → Mary is eliminated**.
- **She → Mary**
  - Mary is feminine and singular.
  - John is masculine, so **She → John is eliminated**.
- **it → ball**
  - "it" is singular and inanimate.

    - "ball" is singular and inanimate, so this is a strong match.
    - "park" is also singular and inanimate, so gender/number alone does not completely eliminate it.
- **him → John**
    - "him" requires a masculine singular antecedent.
    - John fits.
    - Mary does not fit.
    - Dog is grammatically possible because an animal can be referred to as "him" when its sex is known, but there is a semantic problem.
- **they → John + Mary (+ dog)**
    - "they" is plural, so it cannot refer to only John, Mary, or dog individually.
    - It can refer to a group such as **John and Mary** or **John, Mary, and the dog**.

2. Recency

Recency prefers the most recently mentioned compatible entity.

For example:

- Before **He**, the most recent person is **Mary**, but gender agreement eliminates Mary. Therefore **John** is selected.
- Before **She**, John is the most recent person, but gender agreement eliminates John. Therefore **Mary** is selected.
- Before **it**, the recently introduced compatible objects are **ball** and **park**. The ball is closer and is also semantically appropriate for "play with it."

3. Semantic compatibility

The verb and surrounding context provide strong clues.

- **John brought a ball** → the ball is a natural object to **play with**.
- Therefore, **it → ball** is much more appropriate than **it → park**.
- **The dog chased him** → a dog naturally chases a person, so **him → John** is semantically plausible.
- **they all went home** → "all" strongly suggests the complete group: **John + Mary + dog**.

4. Coherence

The resulting entity chains should make the paragraph logically consistent:

- John is introduced → **He** → **him**
- Mary is introduced → **She**
- Ball is introduced → **it**
- John, Mary, and dog form the group → **they all**

Thus, the chains form a coherent story.

c) Constraint graph / table

Variable	Possible domains	Gender/Number	Recency	Semantic compatibility	Final
He	John, Mary	Mary eliminated	John is less recent but compatible	John is a person who can bring a ball	**John**
She	John, Mary	John eliminated	Mary is compatible	Mary can want to play	**Mary**
it	ball, park	Both compatible	Ball is more recent	Playing with a ball is highly suitable	**ball**
him	John, Mary, dog	Mary eliminated	John is recent compatible person	Dog can chase John	**John**
they	John+Mary, John+Mary+dog	Must be plural	Group is recently active	"all" strongly supports complete group	**John+Mary+dog**

Constraint elimination graph

He

├── John ——————————— ✓

└── Mary —— Gender ———— ✗

She

├── John —— Gender ———— ✗

└── Mary ——————————— ✓

it

├── ball —— Recency + Semantic compatibility — ✓

└── park — Recency/semantic preference ————— ✗

him

├── John —— Gender + Semantic compatibility — ✓

├── Mary — Gender ——————————————————— ✗

└── dog —— Less suitable as "him" + coherence — ✗

they

├── John only — Number ————————————————— ✗

├── Mary only — Number ————————————————— ✗

├── John + Mary ————————————————————— ✓

└── John + Mary + dog — "all" + coherence —— ✓✓

d) Final resolved coreference chains

The final chains are:

- **John → He → him**
- **Mary → She**
- **ball → it**
- **John + Mary + dog → they**
- **park** remains an independent entity

- **dog** remains an independent entity until it becomes part of the final plural group referred to by **they**

Rewritten paragraph

John and Mary went to the park. John brought a ball. Mary wanted to play with the ball. The dog chased John excitedly. Finally, John, Mary, and the dog all went home.

This version removes the ambiguous pronouns while preserving the original meaning.

e) Priority order of constraints

A reasonable priority order is:

1. Gender and number agreement → 2. Semantic compatibility → 3. Coherence → 4. Recency

- **Gender and number agreement** should receive high priority because a clear grammatical mismatch can immediately eliminate an antecedent.
- **Semantic compatibility** is next because the meaning of the verb and noun often provides strong evidence. For example, "play with" strongly favors **ball** over **park**.
- **Coherence** ensures that the selected references form a logical entity chain across the paragraph.
- **Recency** is useful as a tie-breaker, but it should not override stronger grammatical or semantic evidence.

What if recency is relaxed?

If the **recency constraint is relaxed**, the system considers older antecedents more freely rather than automatically preferring the nearest compatible entity.

For this paragraph, the main results would remain the same because **gender, semantics, and coherence** provide strong evidence:

- **He → John** because Mary fails gender agreement.
- **She → Mary** because John fails gender agreement.
- **it → ball** because "play with the ball" is semantically much stronger than "play with the park."
- **him → John** because John is the appropriate male/person antecedent.
- **they → John + Mary + dog** because "all" indicates the complete group.

However, without recency, **it → park** or **they → John + Mary** could receive more consideration. Therefore, relaxing recency increases the number of competing candidates and may make the resolution process more computationally expensive and potentially more ambiguous.

2. **Conversation History:** User: “I have an important exam tomorrow but I’m not able to concentrate.”

Required Dialog Act: Advise + Encourage (Give advice and motivate the student).

Constraints:

1. Maintain **entity coherence** (reuse “exam”, “concentrate”, “you”).
2. Use **discourse relations** (Cause-Effect or Elaboration).
3. Include at least **two** of these keywords: focus, break, confident.
4. Response length: 2–3 sentences only.
5. Logical consistency (no contradictory statements).
6. Politeness and positive tone.

Tasks: a) Identify the relevant discourse planning steps needed before generation.

b) Generate **three** different possible responses that satisfy **all** the above constraints.

c) Evaluate the three responses and select the best one. Justify your choice using coherence and constraint satisfaction criteria.

d) Explain how violating any two constraints would affect the quality of the dialog.

a) Relevant discourse planning steps

Before generating the response, the system should perform these steps:

1. **Identify the dialog act:** The response must both **advise** the user and **encourage** them.
2. **Identify key entities:** Maintain the entities **exam**, **concentrate**, and **you** throughout the response.
3. **Determine the discourse relation:** Use a **Cause–Effect** or **Elaboration** relation to connect the advice logically.
4. **Select required content:** Suggest a practical concentration strategy such as taking a short break and reducing distractions.
5. **Add motivation:** Encourage the student to remain **confident** and reassure them that they can prepare effectively.
6. **Check constraints:** Ensure the response has 2–3 sentences, contains at least two keywords (**focus, break, confident**), is polite, positive, and logically consistent.

b) Three possible responses

Response 1

Take a short **break** first, because it can help you regain **focus** and concentrate better on your exam preparation. Stay **confident**, because you still have time to review the most important topics.

Response 2

Since you are finding it difficult to **concentrate**, take a short **break** and then return to your exam preparation with a clear **focus**. You can feel **confident** by reviewing the key topics one at a time.

Response 3

Your exam is important, so remove distractions and take a short **break** to refresh your mind and improve your **focus**. After the break, study the most important topics and stay **confident** in your preparation.

c) Evaluation and best response

Response 2 is the best choice.

- **Dialog act:** It provides direct advice and encouragement.
- **Entity coherence:** It consistently refers to **you**, **concentrate**, and **exam**.
- **Discourse relation:** The word **"Since"** creates a clear cause-effect relationship between difficulty concentrating and taking a break.
- **Keywords:** It contains **concentrate, break, focus, confident**, satisfying the keyword requirement.

- **Length:** It has exactly **2 sentences**.
- **Tone:** It is polite, supportive, and motivating.
- **Logical consistency:** The advice follows a sensible sequence: difficulty concentrating → short break → focused study → confidence.

Therefore, Response 2 provides the strongest combination of **coherence, discourse structure, constraint satisfaction, and encouragement**.

d) Effect of violating two constraints

Violating entity coherence

If the response suddenly changed from **"exam"** and **"concentrate"** to unrelated entities, the connection with the user's original problem would become weak. The response might sound generic rather than directly addressing the student's concern.

Example:
"Take a break and enjoy your favorite movie. Tomorrow will be fine."

This does not maintain a clear connection with the **exam** or the difficulty in **concentrating**.

Violating response length

If the response contained five or six sentences instead of 2–3, it would fail the specified constraint and could become unnecessarily detailed. Since the student is already struggling to concentrate, a short and focused response is more appropriate.

3. **Source Sentence:** "The bank by the river flooded after the storm, but it was saved by quick action."

Note: "bank" is ambiguous (financial institution or riverbank).

Constraints:

1. Correct **Word Sense Disambiguation** using context.
2. Create a **predicate logic** representation (e.g., using predicates like flood(x), location(x,y)).
3. Maintain **discourse structure** (Contrast relation between the two clauses).
4. Preserve all important entities and relations (no information loss).
5. Ensure **text coherence** in the generated output.

Tasks: a) Resolve the ambiguous word "bank" and justify using constraints.

b) Write a predicate logic representation of the sentence.

c) Generate a good **target language sentence** (in any Indian language or simple English paraphrase) that satisfies all constraints.

d) Draw a simple **discourse tree** (RST-style) showing the rhetorical relation.

e) Discuss the advantages of using constraint-based approach over pure statistical machine translation for this sentence

a) Word Sense Disambiguation

The correct meaning of **"bank" is riverbank** (the land beside a river).

Justification:

1. The phrase **"by the river"** strongly indicates that *bank* refers to the side of a river.
2. The predicate **"flooded"** is semantically compatible with a riverbank because a riverbank can be covered by rising water.
3. A **financial bank** cannot naturally be described as "the bank by the river flooded" in this context.
4. Therefore, context, semantic compatibility, and discourse coherence all support the **riverbank** sense.

So:

bank → riverbank (land beside a river)

b) Predicate Logic Representation

Let:

- `bank(b)` = b is a bank/riverbank
- `river(r)` = r is a river
- `storm(s)` = s is a storm
- `action(a)` = a is a quick action
- `location(b,r)` = the bank is located beside the river
- `flood(b)` = the bank flooded
- `cause(s,b)` = the storm caused the flooding
- `quick(a)` = the action was quick
- `save(a,b)` = the action saved the bank

A suitable representation is:

∃b ∃r ∃s ∃a [

river(r) ∧

bank(b) ∧

location(b,r) ∧

storm(s) ∧

flood(b) ∧

cause(s,b) ∧

action(a) ∧

quick(a) ∧

save(a,b)

]

The **contrast relation** between the two clauses can additionally be represented as:

Contrast(

$flood(b) \wedge cause(s,b)$,

$save(a,b) \wedge quick(a)$

)

This captures the idea:

The bank flooded because of the storm, BUT quick action saved it.

c) Target-language sentence

Simple English paraphrase

The riverbank flooded after the storm, but quick action saved the riverbank.

This satisfies the constraints because:

- **“riverbank”** explicitly resolves the ambiguity.
- The **storm** and **quick action** are preserved.
- The flooding and saving relationships are preserved.
- **“but”** maintains the contrast relation.
- The sentence remains coherent and natural.

Tamil version

புயலுக்குப் பிறகு ஆற்றங்கரை வெள்ளத்தில் மூழ்கியது, ஆனால் விரைவான நடவடிக்கையால் ஆற்றங்கரை காப்பாற்றப்பட்டது.

Meaning: *After the storm, the riverbank was flooded, but the riverbank was saved through quick action.*

d) Simple RST-style discourse tree

```
            Sentence
               |
             CONTRAST
            /       \
           /         \
   Situation/Problem    Resolution
```

```
        |              |
   The riverbank flooded   Quick action
     after the storm      saved it
```

More detailed:

```
              CONTRAST
             /        \
            /          \
     Clause 1          Clause 2
       |                  |
   Bank flooded         Bank was saved
   after storm          by quick action
```

The **Contrast** relation is important because the first clause describes a negative event (**flooding**), while the second clause presents an opposing positive outcome (**being saved**).

e) Advantages of a constraint-based approach over pure statistical machine translation

1. **Better ambiguity resolution:** The system uses "by the river" and "flooded" to correctly select **riverbank** rather than financial institution.
2. **Preserves logical meaning:** Predicate constraints ensure that important relations such as **bank–river**, **storm–flood**, and **action–saving** are retained.
3. **Maintains discourse structure:** The **Contrast** relation represented by "but" is explicitly preserved.
4. **Reduces semantic errors:** A purely statistical system might select an incorrect sense of "bank" based on frequent patterns, whereas constraints require the selected meaning to fit the context.
5. **Improves coherence:** The same entity is maintained across both clauses, preventing inconsistent references.
6. **More interpretable:** Constraint-based decisions can be explained: *"bank" was selected as riverbank because it is located by a river and can flood.*
7. **Useful for low-resource languages:** Explicit linguistic and logical constraints can help when large amounts of training data are unavailable.

Overall, a constraint-based approach is especially useful for this sentence because it combines lexical meaning, semantic compatibility, logical relations, and discourse structure rather than relying only on statistical probability.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________